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Correlates of Poverty: Evidence from the Community-Based Monitoring System (CBMS) Data

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This study identified correlates of poverty for Pasay City and Mogpog, Marinduque representing an urban and a rural area in the Philippines, respectively, by utilizing the 2005 census data from its Community-Based Monitoring System. Regression models with the arcsine of the square root of barangay level poverty incidence as dependent variable were investigated which allowed the identification of the correlates of poverty at the barangay level. Results showed that the significant correlates of barangay poverty incidence were average household size, proportion of households whose housing units/lot are not owned and proportion of households who own telephone/cellphone. Furthermore, lower poverty incidences were observed in barangays located in an urban area.

Keywords: poverty, poverty incidence, correlates of poverty, community-based monitoring system

For purposes of eradicating poverty, information is required about the poor and their circumstances – who they are, where they live, what social and economic conditions they face, how they respond to programs and projects intended for them, and so forth. With such poverty profile, Local Government Units (LGUs) can prepare comprehensive development plans geared towards poverty reduction, among others. Through its improved design and

implementation, benefits of poverty alleviation efforts are maximized. A very vital tool in this undertaking is statistical data.

The National Statistics Office (NSO) is the main source of official statistics in the Philippines that LGUs and policy makers relied on when they formulate policies and implement projects and programs geared towards poverty reduction. With the Family Income and Expenditure Survey (FIES) they conduct every three years, poverty statistics

are being computed down to the provincial level with an acceptable measure of reliability.

However, poverty statistics at smaller geographical division such as municipalities/cities and barangays are needed to better implement and monitor poverty reduction strategies. Although small area estimation methodologies are currently being employed that utilizes information from surveys and censuses, alternative approaches to generating small area poverty statistics is needed that utilizes baseline information that will complement the official poverty statistics being generated by NSO.

Objectives of the Study

The primary objective of this study was to determine correlates of poverty, utilizing the data from the Community-Based Monitoring System (CBMS) for effective beneficiary targeting. Specifically, this study aimed to (1) generate poverty incidence at the barangay level using the CBMS data set from Pasay City and Mogpog, Marinduque, an urban and a rural area, respectively; and (2) determine the correlates of poverty and generate poverty models useful for monitoring purposes in these areas.

Generated poverty incidences can be used in determining the poverty situation at the barangay level which can aid LGUs in their local level program and policy development. Coupled with identified correlates of poverty, LGUs' ability to target the most needy can be enhanced.

Furthermore, such correlates of poverty can be used for evaluating existing programs and projects that will determine if they work or not, and will indicate what changes are needed for better utilization of resources and targeting of beneficiaries.

METHODOLOGY

The study utilized CBMS data collected from Pasay City and Mogpog, Marinduque in the Philippines during the 2005 census year. The

data were a complete enumeration of 65,117 households in 201 barangays in Pasay City, and 6,650 households in 38 barangays in Mogpog, Marinduque.

Variables included in the study were those that are available across LGUs of Pasay City and Mogpog, Marinduque. It included demographic and socio-economic characteristics of the households from which several indices were created to facilitate the model-building process.

Using the reported total income of the household, a household's per capita income was computed as the household's total income divided by the size of the household. The poverty status of the household, whether poor or non-poor, was then determined following the official methodology of the Philippines for poverty measurement. That is, a household was classified as poor if its per capita income is below the poverty threshold, the per capita income necessary to meet basic food and non-food needs of a household. Otherwise, the household was classified as non-poor (National Statistical Coordination Board [NSCB], 2006). For 2005, NSCB estimated the poverty thresholds for Pasay City at Php 18,723 and for Marinduque at Php 13,859.

With the barangay as small area unit, the poverty incidences were estimated by taking the ratio of the total number of poor households to the total number of households in that barangay. In the model-building process, this poverty incidence at the barangay level was considered as the variable whose values are to be predicted. The independent variables or predictors in the model were the indices constructed based on the available variables.

In generating poverty profiles for Pasay City and Mogpog, Marinduque, bivariate analyses were performed that allowed the characterization of the poor. Because some of the variables were in the nominal scale such as gender of the household head, while others were in at least ordinal scale such as size of the household, both contingency and correlation analyses were performed at 5% level of significance to identify the correlates of poverty.

Multivariate analysis was likewise performed to determine determinants of barangay poverty incidence. This regression model of the poverty situation at the barangay level allowed the investigation of the effects of a particular aggregated correlate of poverty conditional on the levels of the other identified correlates.

Because the response variable were proportions, the regression model was specified as

$$\arcsin(\sqrt{Y_i}) = \beta X_i + \varepsilon_i$$

where Y_i is the poverty incidence of the i^{th} barangay, β is the vector of regression coefficients, X_i is the vector of indices based on the identified correlates of poverty, and ε_i is the regression model noise. The arcsine transformation allows proportions to be made nearly “normal”.

Residual analyses were performed to determine if underlying assumptions of the linear regression model were satisfied. Assumptions investigated included normality with the use of the Shapiro-Wilk’s test statistic, independence with the use of the Durbin-Watson test statistic, and absence of multicollinearity among independent variables with the use of the tolerance and variance inflation statistics.

All results were generated using the Statistical Analysis System (SAS) software.

RESULTS AND DISCUSSION

Poverty Incidence. Poverty incidence indicated that 101 of the 201 barangays in Pasay City had poverty incidence estimates of at most 10.0% (Table 1). The distribution of poverty incidence in Pasay City was skewed to right, with only 3.5% of the barangays having poverty incidences greater than 30%. The poorest barangay had a poverty incidence of 42.1%, while one barangay had no poor household.

On the other hand, all of the 36 barangays in Mogpog, Marinduque had poverty incidence estimates higher than 20%. This was expected since Mogpog, Marinduque is classified as a rural area and studies have shown that location attributes (whether urban or rural) determine living standard of households (Balisacan, 1997; Albert & Collado, 2004). In particular, households located in urban areas were better-off relative to those in rural areas. Hence, the distribution of poverty incidence was observed to be skewed to the left, with nearly 70% of the barangays having poverty incidences greater than 60%. Also, the poorest barangay had a poverty incidence as high as 93.62%. That is, nine out of 10 households in this barangay were poor. The least poverty incidence among the barangays in Mogpog, Marinduque was 28.22%.

Table 1

Percentage distribution of poverty incidence by area, 2005

POVERTY INCIDENCE (%)	AREA	
	Pasay City	Mogpog, Marinduque
<= 10	50.3	0.0
11 – 20	32.3	0.0
21 – 30	13.9	2.8
31 – 40	3.0	11.1
41 – 50	0.5	11.1
51 – 60	0.0	5.6
61 – 70	0.0	27.8
71 – 80	0.0	16.7
81 – 90	0.0	19.4
> 90	0.0	5.6

Correlates of Poverty. Several household demographic characteristics are significant indicators of higher poverty incidence in both Pasay City and Mogpog, Marinduque: “male household head”, “without OFW”, and “with handicapped” (Table 2). There was also a very evident trend wherein the lower the household head’s educational level, the higher the poverty incidence.

From Table 3, engagement in the following economic activities was found to be significant indicators of poverty incidence for both Pasay City and Mogpog, Marinduque: engagement in crop farming/gardening, livestock, transportation/storage/communication, and construction. But other economic characteristics of households were found to be significant on only one area since some data were not available for the other area.

Table 2

Distribution of poor households by demographic characteristics and area

VARIABLE	CATEGORIES	PASAY CITY		MOGPOG, MARINDUQUE	
		% POOR	P-VALUE	% POOR	P-VALUE
SEX (Sex of Household Head)	Female	9.2	< 0.0001	54.0	< 0.0001
	Male	14.9		69.3	
EDUC (Highest educational attainment of household head)	No grade completed	30.3	< 0.0001	76.1	< 0.0001
	Elementary undergraduate	24.6		77.1	
	Elementary graduate	22.4		74.5	
	High school undergraduate	24.3		71.5	
	High school graduate	14.2		62.7	
	College undergraduate	9.6		46.9	
	College graduate and beyond	3.7		27.1	
OFWIND (OFW indicator)	Without OFW	14.4	< 0.0001	69.0	< 0.0001
	With OFW	5.0		23.3	
UNIPARENT (Single-parent indicator)	Without single parent	13.6	0.3873	66.0	0.0517
	With single parent	14.0		70.7	
THIRDSEX (Third sex indicator)	Without third sex	13.6	0.0006	N.A.	N.A.
	With third sex	18.2		N.A.	
HANDICAPPED (Handicapped indicator)	Without handicapped	13.4	< 0.0001	65.7	0.0022
	With handicapped	22.2		71.7	
BOARDIND (Boarder / bedspacer Indicator)	Without boarder / bedspacer	13.2	< 0.0001	N.A.	N.A.
	With boarder / bedspacer	9.0		N.A.	
ORGIND (Member of any community organization)	Not a member	N.A.	N.A.	69.2	< 0.0001
	Member			55.9	

N.A. - not available.

Table 3*Distribution of poor households by economic activities and area*

VARIABLE	CATEGORIES	PASAY CITY		MOGPOG, MARINDUQUE	
		% POOR	P-VALUE	% POOR	P-VALUE
JOBIND (Job/work indicator)	Without	13.3	0.1118	59.5	< 0.0001
	With	13.7		68.7	
CROPIND (Engaged in crop farming and gardening)	Not engaged	13.7	< 0.0001	63.9	< 0.0001
	Engaged	7.3		70.5	
POULTIND (Engaged in livestock / poultry)	Not engaged	13.6	0.0300	59.7	< 0.0001
	Engaged	7.5		71.9	
FISHIND (Engaged in fishing)	Not engaged	13.6	0.3320	62.1	< 0.0001
	Engaged	17.6		82.7	
FORIND (Engaged in forestry)	Not engaged	13.6	0.1757	64.4	< 0.0001
	Engaged	7.8		72.3	
SALIND (Engaged in wholesale / retail)	Not engaged	13.6	0.5569	68.5	< 0.0001
	Engaged	13.8		57.6	
PUBLIND (Engaged in publishing)	Not engaged	13.6	0.1143	N.A.	N.A.
	Engaged	9.1			
MANIND (Engaged in manufacturing)	Not engaged	13.6	0.0768	66.2	0.1806
	Engaged	11.0		70.6	
MAINTIND (Engaged in maintenance services)	Not engaged	13.6	0.0296	N.A.	N.A.
	Engaged	16.0			
FOODIND (Engaged in food services)	Not engaged	13.6	0.0086	N.A.	N.A.
	Engaged	10.2			
ENTERTAIN (Engaged in entertainment services)	Not engaged	13.6	0.0009	N.A.	N.A.
	Engaged	7.0			
SERVIND (Engaged in community, social & personal services)	Not engaged	13.6	0.9337	66.5	< 0.0001
	Engaged	13.5		43.8	
COMPUTIND (Engaged in computer communication)	Not engaged	13.6	0.0012	N.A.	N.A.
	Engaged	6.9			
TRNIND (Engaged in transpo, storage & comm.)	Not engaged	13.3	< 0.0001	67.1	< 0.0001
	Engaged	17.9		59.0	
MININD (Engaged in mining & quarrying)	Not engaged	13.6	0.7513	65.9	< 0.0001
	Engaged	14.8		85.2	
CNSIND (Engaged in construction)	Not engaged	13.5	< 0.0001	65.7	0.0018
	Engaged	19.2		71.8	
EOTHIND (Engaged in other activities NEC)	Not engaged	13.6	0.0509	66.3	0.8414
	Engaged	16.7		65.3	

N.A. - not available.

Table 4*Distribution of poor households by household's basic needs and area*

VARIABLE	CATEGORIES	PASAY CITY		MOGPOG, MARINDUQUE	
		% POOR	P-VALUE	% POOR	P-VALUE
TENUR (Tenure status of house / lot)	Owner, owner-like ... house and lot	9.5	< 0.0001	57.2	< 0.0001
	Rent house / room including lot	15.6		52.9	
	Own house / rent lot	14.2		73.5	
	Own house /rent-free lot with consent	17.1		78.7	
	Own house /rent-free lot without consent	25.2		82.6	
	Rent-free hse and lot with consent	13.6		73.6	
	Rent-free hse and lot without consent	28.9		86.7	
	Other	43.3		72.7	
HTYPE (House type)	Single house	13.2	< 0.0001	N.A.	N.A.
	Duplex	15.6			
	Apartment . . .	10.5			
	Commercial . . .	12.2			
	Other	36.6			
WALL (Construction materials of walls)	Strong	10.3	< 0.0001	54.4	< 0.0001
	Light	26.3		83.0	
	Salvaged / makeshift	30.0		88.3	
	Mixed but predominantly strong	20.0		67.0	
	Mixed but predominantly light	32.2		81.2	
	Mixed but predominantly salvaged	40.7		85.4	
ROOF (Construction materials of roof)	Strong	10.6	< 0.0001	58.2	< 0.0001
	Light	18.9		83.4	
	Salvaged / makeshift	19.2		89.7	
	Mixed but predominantly strong	16.4		74.9	
	Mixed but predominantly light	32.1		82.4	
	Mixed but predominantly salvaged	37.3		85.4	
WATER (Type of water facility)	Community water system-own	33.3	< 0.0001	51.6	< 0.0001
	Community water system – shared	24.0		75.2	
	Deep well – own	19.2		34.5	
	Deep well – shared	26.5		66.2	
	Artesian well – own	25.0		41.0	
	Artesian well – shared	27.3		68.9	
	Dug/shallow well – own	22.2		72.5	
	Dug/shallow well – shared	14.3		9.1	
	River, stream, lake, ...	0.0		79.8	
	Bottled water	9.1		8.9	
	Others	28.1		58.1	
TOIL (Type of toilet facility)	Water-sealed. . . own	6.7	< 0.0001	55.7	< 0.0001
	Water-sealed. . . – shared	18.9		70.2	
	Closed pit	11.6		81.6	
	Open pit	19.3		80.3	
	No toilet	34.3		87.2	
	Others	33.1		83.3	
FSHORT (Experienced food shortage)	Without food shortage	13.2	< 0.0001	64.5	< 0.0001
	With food shortage	46.7		88.2	

N.A. - not available.

A lower poverty incidence was observed among households in Pasay City who were engaged in crop farming and gardening, livestock/poultry, food services, entertainment services, and computer communication. On the other hand, a higher poverty incidence was seen among households who were engaged in maintenance services, transportation/storage/communication, and construction.

This is contrary to the findings from Mogpog, Marinduque where higher poverty incidences were observed among households engaged in crop farming and gardening, livestock/poultry, fishing, forestry, transportation, storage and communication, mining and quarrying, and construction. While the following economic

activities were significant indicators of lower poverty incidence: wholesale/retail, and community/social/personal services.

Households' basic needs such as tenure status of house and lot, construction materials of wall and roof, types of water and toilet facilities, and experience of food shortage were significant predictors of the poverty status of households for both Pasay City and Mogpog, Marinduque (Table 4).

Ownership of household consumer durables such as TV, VHS, computer, refrigerator, electric iron, LPG/gas stove/range, washing machine, microwave oven, telephone/cellphone, aircon, and vehicles was significantly associated with poverty incidence (Table 5). Specifically, poverty

Table 5

Distribution of poor households by household's consumer durables and area

VARIABLE	CATEGORIES	PASAY CITY		MOGPOG, MARINDUQUE	
		% POOR	P-VALUE	% POOR	P-VALUE
TV (Own TV)	Without	22.1	< 0.0001	81.9	< 0.0001
	With	12.3		53.6	
VHS (Own VHS / VCD / DVD player)	Without	20.5	< 0.0001	77.4	< 0.0001
	With	9.7		48.5	
COMPUTER (Own computer)	Without	15.4	< 0.0001	68.0	< 0.0001
	With	4.1		20.8	
REF (Own ref)	Without	20.7	< 0.0001	77.8	< 0.0001
	With	7.9		35.3	
IRON (Own electric iron)	Without	22.9	< 0.0001	80.7	< 0.0001
	With	10.7		47.5	
STOVE (Own LPG / gas stove / range)	Without	23.2	< 0.0001	79.2	< 0.0001
	With	10.8		46.1	
WMACH (Own washing machine)	Without	18.0	< 0.0001	73.0	< 0.0001
	With	8.5		32.4	
MICROW (Own microwave oven)	Without	15.1	< 0.0001	68.4	< 0.0001
	With	5.5		20.1	
PHONE (Own telephone / cell phone)	Without	25.3	< 0.0001	78.1	< 0.0001
	With	8.4		43.9	
AIRCON (Own aircon)	Without	15.0	< 0.0001	67.4	< 0.0001
	With	4.2		20.1	
CAR (Own vehicles)	Without	14.7	< 0.0001	70.7	< 0.0001
	With	5.1		32.0	

incidence was found to be much higher among households without these household consumer durables.

Access to government or private organization programs on gender issues, peace and order, health, education, credit, and other programs were found to be significant predictors of the poverty status of households residing in Pasay City (Table 6). A higher poverty incidence can be observed among households that received these programs compared to the households that did not receive them. Effect of government or private organization programs on the poverty status of household in Mogpog, Marinduque was not

investigated since similar data were not available for Mogpog, Marinduque.

Poverty Incidence Model. The Pasay City poverty incidence regression model was found to be overall significant ($p < 0.0001$) with 18 significant correlates of poverty incidence enumerated in Table 7. Correlates which positively or directly affect barangay poverty incidence in Pasay City include AVE_HSIZE, PROP_SEX, PROP_OFWIND, PROP_THIRDSEX, PROP_FOODIND, PROP_TENUR, PROP_WALL, PROP_WMACH, PROP_AIRC, and PROP_OTHPRIND. On the other hand, correlates which negatively or

Table 6

Distribution of poor households by household's availment of programs, Pasay City

VARIABLE	CATEGORIES	% POOR	P-VALUE
SEXPROG (Received programs on gender issues)	Did not receive	13.5	< 0.0001
	Received	18.9	
PEACEPROG (Received programs on peace and order)	Did not receive	13.3	0.0004
	Received	14.3	
MEDHEAL (Received programs on health)	Did not receive	13.5	0.0164
	Received	14.2	
MSCHOL (Received programs on education)	Did not receive	13.5	< 0.0001
	Received	26.3	
MTRAININD (Received programs on livelihood training)	Did not receive	13.7	0.3034
	Received	11.8	
ASSHLOTIND (Received programs on housing)	Did not receive	13.7	0.1992
	Received	11.1	
CREDIND (Received credit programs)	Did not receive	13.6	0.0020
	Received	19.0	
CLEANPROG (Received cleanliness programs)	Did not receive	13.7	0.7314
	Received	13.5	
JOBPROG (Received employment programs)	Did not receive	13.6	0.2311
	Received	15.7	
OTHPRIND (Received other programs)	Did not receive	13.3	< 0.0001
	Received	17.9	

inversely affect barangay poverty incidence in Pasay City include PROP_BOARDIND, PROP_JOBIND, PROP_COMPUTIND, PROP_COMPUTER, PROP_REF, PROP_PHONE, and PROP_CLEANPROG.

There was no multicollinearity among the significant correlates under this poverty incidence

model as supported by the tolerance and variance inflation statistics. Furthermore, this Pasay City poverty incidence model also satisfied normality ($p = 0.9600$), and consequently homoscedascity, as well as independence assumption of the residuals ($p < DW = 0.7531$ and $p > DW = 0.2469$). The adjusted R-square of this poverty incidence

Table 7

Significant correlates in Pasay City poverty incidence model

CORRELATES	PARAMETER ESTIMATE*	P-VALUE
AVE_HSIZE (Average household size)	0.10068 (0.01189)	<0.0001
PROP_SEX (Proportion of households who are headed by a male person)	0.28332 (0.07901)	0.0004
PROP_OFWIND (Proportion of household with an OFW member)	0.43122 (0.12462)	<0.0007
PROP_THIRDSEX (Proportion of households with members belonging to the third sex)	1.46496 (0.46236)	0.0016
PROP_BOARDIND (Proportion of households with boarders/bedspacers)	-1.03444 (0.25858)	<0.0001
PROP_JOBIND (Proportion of household heads who are employed)	-0.12098 (0.06091)	0.0485
PROP_FOODIND (Proportion of households engaged in food services)	0.64818 (0.24813)	0.0097
PROP_COMPUTIND (Proportion of households engaged in computer communication)	-1.91134 (0.66630)	0.0046
PROP_CNSIND (Proportion of households engaged in construction)	-0.67962 (0.16586)	<0.0001
PROP_TENUR (Proportion of households whose housing units/lot are not owned)	0.18870 (0.02890)	<0.0001
PROP_WALL (Proportion of households living in housing units with walls made of light or salvaged/makeshift materials)	0.18761 (0.05066)	0.0003
PROP_COMPUTER (Proportion of households who own computer)	-0.22789 (0.08066)	0.0052
PROP_REF (Proportion of households who own refrigerator)	-0.45935 (0.06918)	<0.0001
PROP_WMACH (Proportion of households who own washing machine)	1.6865 (0.06417)	0.0093
PROP_PHONE (Proportion of households who own telephone/cellphone)	-0.27615 (0.06104)	<0.0001
PROP_AIRC (Proportion of households who own aircon)	0.31213 (0.08313)	0.0002
PROP_CLEANPROG (Proportion of households who received cleanliness programs)	-0.05871 (0.01855)	0.0018
PROP_OTHPRIND (Proportion of households who received other programs)	0.13592 (0.03523)	0.0002

* Numbers inside parenthesis are standard error of the estimate.

Table 8*Significant correlates in Mogpog poverty incidence model*

CORRELATES	PARAMETER ESTIMATE*	P-VALUE
AVE_HSIZE (Average household size)	0.12249 (0.04331)	0.0081
PROP_CROPIND (Proportion of households engaged in crop farming and gardening)	-0.19126 (0.06606)	0.0069
PROP_TV (Proportion of households who own TV)	-0.58926 (0.21388)	0.0097
PROP_IRON (Proportion of households who own electric iron)	-0.42589 (0.17283)	0.0195

* Number inside parenthesis are standard error of the estimate.

Table 9*Significant correlates in the poverty incidence model for the combined data of Pasay City and Mogpog, Marinduque*

CORRELATES	PARAMETER ESTIMATE*	P-VALUE
AREA (Location of the barangay)	-0.37893 (0.02167)	<0.0001
AVE_HSIZE (Average household size)	0.10355 (0.01133)	<0.0001
PROP_TENUR (Proportion of households whose housing units/lot are not owned)	0.23787 (0.03569)	<0.0001
PROP_PHONE (Proportion of households who own telephone/cellphone)	-0.65644 (0.04200)	<0.0001

* Number inside parenthesis are standard error of the estimate.

model was 84.77% indicating that 84.77% of the variation in poverty incidence at the barangay level is accounted for by the significant correlates adjusted for the number of correlates used.

On the other hand, the significant Mogpog poverty incidence regression model ($p < 0.0001$) had four correlates of poverty incidence having an adjusted R-square of 79.83%. The four significant correlates are enumerated in Table 8.

A positive relationship was observed between poverty incidence and AVE_HSIZE. This implied that barangays with high average household

sizes have higher poverty incidences. On the other hand, inverse relationships were observed between poverty incidence and the following correlates: PROP_CROPIND, PROP_TV, and PROP_IRON. This implied that the higher the proportions of households who own TV, who own electric iron, and who engage in crop farming or gardening in a barangay, the lower the poverty incidence in this barangay.

This Mogpog poverty incidence model passed the diagnostic checking performed on its residuals. Tolerance and variance inflation

statistics indicated absence of multicollinearity. It satisfied assumptions of normality ($p = 0.5294$) and independence of residuals ($p < DW = 0.6055$ and $p > DW = 0.3945$).

Multiple regression analysis was likewise performed on the combined CBMS data collected from Pasay City and Mogpog, Marinduque. Results showed that the model was significant ($p < 0.0001$) with four correlates that are not multicollinear (Table 9). This model has an adjusted R-square of 91.40% satisfying normality ($p = 0.2540$) and independence ($p < DW = 0.3287$ and $p > DW = 0.6713$) of residuals assumptions.

An inverse relationship was observed between barangay level poverty incidence and location of the barangay, that is, lower poverty incidences are observed in barangays located in a rural area. Also, the proportion of households who own telephone/cellphone negatively affect barangay level poverty incidence. This implied that the higher the proportions of households who own telephone/cellphone in a barangay, the lower the poverty incidence is in that barangay. On the other hand, average household size and proportion of households whose housing units/lot are not owned positively affect barangay level poverty incidence.

CONCLUSIONS

Utilizing the 2005 CBMS census data, this study identified correlates of poverty for Pasay City and Mogpog, Marinduque, selected urban and rural areas in the Philippines, respectively. The results from the bivariate and multivariate analyses showed that for both areas, average household size positively affect barangay level poverty incidence. It is imperative that family planning programs be aggressively disseminated to properly educate

couples and adults on responsible parenthood and on the direct correlation between large household size and higher poverty incidence.

Further analyses on the combined CBMS data from both areas allowed the assessment of the effect of the location of a barangay (whether urban or rural) on the barangay level poverty incidence. Results showed that lower poverty incidences were observed in barangays located in an urban area, that is, households located in urban areas were better-off relative to those in rural areas. Barangays having higher proportions of households who own telephone/cellphone in a barangay had lower poverty incidences, while barangays with higher proportions of households whose housing units/lot are not owned had higher poverty incidences. Hence, appropriate housing programs should be studied and implemented in order to achieve poverty reduction.

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